Strategic Fingerprints Under the Microscope: Quantifying Prompt Sensitivity in LLM-Played Iterated Prisoner's Dilemma

Robert Wang and Charlotte Roth

Algorithmic Game Theory, Spring 2026

1. Introduction

As large language model agents are increasingly deployed in competitive real-world settings such as algorithmic trading, automated negotiation, and multi-agent AI systems, understanding their emergent strategic behavior is no longer a purely academic question. A model that cooperates too readily gets exploited; one that defects too aggressively destabilizes productive equilibria. Characterizing these tendencies, and understanding what controls them, is critical for the safe deployment of LLM agents in adversarial environments.

Classical results from human-designed Prisoner's Dilemma experiments, most notably Robert Axelrod's tournament, showed that simple cooperative strategies like Tit-for-Tat dominated over complex ones. Whether LLM agents, which bring fundamentally different reasoning capabilities and training objectives, replicate or diverge from these results is an open and practically important question.

Recent work has begun to explore this space. Payne & Alloui-Cros (2025) conducted the first evolutionary IPD tournament pitting LLM agents from OpenAI, Google, and Anthropic against classical strategies, finding that each model exhibits a distinctive and persistent "strategic fingerprint." They described these qualitatively: Gemini as a "calculating game theorist," OpenAI as a "principled idealist," and Claude as a "sophisticated diplomat." Willis et al. (2025) showed that different LLMs exhibit distinct biases when generating full IPD strategies, with prompting disposition affecting whether cooperative or exploitative equilibria emerge. Vallinder & Hughes (2024) studied cultural evolution of cooperation in LLM agent societies across generations. Warnakulasuriya et al. (2025) demonstrated that LLM agents in n-player social dilemmas can replicate cooperation dynamics predicted by traditional mathematical models.

Our project builds on this emerging literature and extends it in two concrete directions. First, we treat prompt framing as a systematic experimental variable rather than a fixed methodological choice — testing whether the same model produces different strategic behavior under neutral, rational, and cooperative framings. Second, we operationalize the qualitative "strategic fingerprint" concept into a set of quantitative behavioral metrics that enable rigorous comparison across models, framings, and classical baselines.

2. Background

2.1 The Prisoner's Dilemma

The Prisoner's Dilemma is a two-player game in which each player independently chooses to either cooperate or defect. The payoff structure creates a tension between individual and collective rationality: mutual cooperation yields a moderate reward for both players (R = 3), mutual defection yields a lower punishment (P = 1), but a lone defector earns the highest temptation payoff (T = 5) while the cooperator receives the sucker's payoff (S = 0). This satisfies the standard ordering T > R > P > S with 2R > T + S, ensuring that mutual cooperation is collectively optimal while individual defection is always tempting.

In the iterated version, players interact repeatedly over multiple rounds, allowing strategies to condition on the history of past play. This introduces the possibility of reciprocity, retaliation, and forgiveness — dynamics absent from the single-shot game.

2.2 Axelrod's Tournament

Robert Axelrod's computer tournaments in the early 1980s established foundational results about iterated Prisoner's Dilemma strategy. The winning strategy, Tit-for-Tat (TFT), submitted by Anatol Rapoport, had just two rules: cooperate on the first move, then mirror the opponent's previous action. Axelrod identified three properties of successful strategies: they are "nice" (never defect first), "retaliatory" (punish defection quickly), and "forgiving" (resume cooperation after retaliation). These properties form the conceptual basis for our behavioral fingerprint metrics.

2.3 Classical Strategies

Our tournament includes nine classical strategies spanning the cooperation-defection spectrum. Always Cooperate and Always Defect represent the extremes. Tit-for-Tat mirrors the opponent's last move. Pavlov (Win-Stay-Lose-Shift) repeats its action after a favorable outcome and switches otherwise. Grim Trigger cooperates until any defection, then defects permanently. Generous Tit-for-Tat adds a probability of forgiving defection. We also include three strategies that create more nuanced strategic pressure: Joss (TFT but defects approximately 10% randomly), Suspicious TFT (TFT but defects on the first move), and Tester (probes opponents for exploitability through strategic early defection).

2.4 Evolutionary Game Theory

Evolutionary game theory extends classical game theory by asking not just which strategies are rational, but which strategies survive competition over time. The replicator equation models population dynamics: strategies earning above-average payoffs grow in frequency, while those earning below-average payoffs shrink. A strategy is an Evolutionarily Stable Strategy (ESS) if a population playing that strategy cannot be invaded by any small group of mutants playing a different strategy. We use both continuous-time replicator dynamics and finite-population Moran processes to analyze the evolutionary properties of our tournament results.

2.5 Shadow of the Future

We use probabilistic termination ("shadow of the future") rather than fixed-length matches. After each round, there is a fixed probability $p = 0.10$ that the game ends, with a hard cap of 100 rounds. This eliminates the backward induction problem inherent in fixed-length games and creates incentives for sustained cooperation. This design follows Payne & Alloui-Cros (2025), enabling direct comparison with their results.

3. Methodology

3.1 Experimental Design

We conduct a round-robin tournament between 15 strategies: 9 classical strategies implemented via the axelrod-python library (Knight et al., 2016) and 6 LLM strategies (2 models × 3 prompt framings). Each strategy pair plays 5 replicate matches with probabilistic termination ($p = 0.10$, cap at 100 rounds). The resulting payoff matrix feeds into replicator dynamics (via nashpy), ESS invasion testing, and Moran process simulations.

3.2 LLM Strategy Implementation

We test two frontier-class, non-reasoning models: Claude Sonnet 4.6 (Anthropic) and GPT-4o (OpenAI). We deliberately do not enable extended thinking on Claude Sonnet 4.6 — which supports it as an optional mode — to maintain experimental symmetry with GPT-4o, which is a standard (non-reasoning) model. This ensures behavioral differences reflect model training rather than reasoning-mode artifacts.

Each LLM strategy operates by receiving the full game history (all previous rounds with actions and payoffs for both players, plus running score totals) and returning a single-word response: COOPERATE or DEFECT. If the API call fails or the response cannot be parsed, the strategy defaults to COOPERATE. Parse failure rates were below 0.13% across all strategies, confirming that this fallback did not materially affect results.

3.3 Prompt Framing Conditions

All three framings share the same core game description (payoff structure, termination mechanism, response format). They differ only in a brief persona preamble:

Neutral: No persona — the bare game description only.

Rational: "You are a rational economic agent. You make decisions to maximize your expected payoff using game-theoretic reasoning."

Cooperative: "You are a cooperative partner. You value building mutual trust and achieving good outcomes for both parties."

3.4 Behavioral Fingerprint Metrics

We operationalize strategic behavior through six quantitative metrics computed from game histories:

Cooperation rate: Overall fraction of moves that were COOPERATE.

First-move cooperation: Fraction of games where the strategy cooperated on round 1.

Retaliation rate: P(DEFECT | opponent defected last round). Measures responsiveness to exploitation.

Forgiveness rate: P(COOPERATE | mutual defection last round). Measures willingness to de-escalate after conflict.

Opportunism rate: P(DEFECT | opponent cooperated last round). Measures tendency to exploit cooperative opponents.

Good partner rate: P(COOPERATE | opponent cooperated last round). Measures reliability as a cooperative partner.

These metrics extend Axelrod's qualitative categories (nice, retaliatory, forgiving) into continuous, measurable quantities that enable precise cross-strategy comparison.

3.5 Evolutionary Analysis

We analyze evolutionary dynamics through three complementary methods. Replicator dynamics (via nashpy): continuous-time ODE integration from a uniform initial distribution across all 15 strategies, run for 500 generations. Replicator-mutation dynamics: the same replicator dynamics with a 1% symmetric mutation rate, preventing strategies from going to exactly zero. Moran process: 50 independent runs of a finite-population (N=50) stochastic birth-death process. ESS invasion testing: for each strategy, we initialize a population with 95% incumbent and 5% invader and run replicator dynamics to test stability.

3.6 Infrastructure and Reproducibility

Tournament infrastructure uses axelrod-python (Knight et al., 2016), a peer-reviewed library providing canonical implementations of classical IPD strategies. Evolutionary dynamics use nashpy (Knight, 2018) for replicator dynamics, mutation dynamics, and Nash equilibrium computation. Our novel contributions — LLM strategy integration, prompt framing conditions, and behavioral fingerprint metrics — are implemented in custom Python code. All experiments use a fixed random seed for reproducibility.

4. Results

4.1 Behavioral Fingerprints: Model-Level Differences

The fingerprint metrics reveal clear, persistent differences between Claude and GPT that hold across all three framing conditions.

Claude exhibits what we term a "diplomatic cooperator" profile: high cooperation (0.824–0.913), reliable but not extreme retaliation (0.926–0.966), moderate forgiveness (0.048–0.099), and very low opportunism (0.009–0.015). Claude almost never exploits a cooperative opponent, retaliates against defection but not maximally, and occasionally forgives.

GPT exhibits a "rigid retaliator" profile: cooperation varies widely by framing (0.728–0.929), but retaliation is consistently near-perfect (0.904–0.992). GPT punishes defection harder and forgives less than Claude in almost every condition.

Metric	Claude (range)	GPT (range)
Cooperation rate	0.824 – 0.913	0.728 – 0.929
Retaliation rate	0.926 – 0.966	0.904 – 0.992
Forgiveness rate	0.048 – 0.099	0.009 – 0.278
Opportunism rate	0.009 – 0.015	0.008 – 0.028

Figure 1: Behavioral fingerprint metrics across all 15 strategies. LLM strategies show distinct profiles from classical baselines.

4.2 Prompt Framing Sensitivity

Our primary novel finding is that prompt framing produces measurable behavioral shifts, but the magnitude and direction of these shifts differ substantially between models.

Claude's framing sensitivity is narrow (8.9pp cooperation range):

Framing	Coop. Rate	Retaliation	Forgiveness	Opportunism
Neutral	0.835	0.956	0.062	0.015
Rational	0.913	0.966	0.048	0.009
Cooperative	0.824	0.926	0.099	0.013

Counterintuitively, Claude's rational framing produces the highest cooperation rate (0.913) and lowest opportunism (0.009). This suggests that Claude under rational framing reasons its way to cooperation as the payoff-maximizing strategy in iterated games. The cooperative framing produces the highest forgiveness (0.099) and the lowest retaliation (0.926) of the three Claude variants, indicating that the cooperative persona softens Claude's defensive behavior, making it more tolerant of exploitation and quicker to attempt reconciliation after conflict. Interestingly, this does not translate into the highest overall cooperation rate (0.824 vs. Rational's 0.913), suggesting that prompt framing operates on the quality of strategic behavior (how the model handles conflict) rather than simply the quantity of cooperation.

GPT's framing sensitivity is dramatic (20.1pp cooperation range):

Framing	Coop. Rate	Retaliation	Forgiveness	Opportunism
Neutral	0.728	0.985	0.017	0.028
Rational	0.806	0.992	0.009	0.021
Cooperative	0.929	0.904	0.278	0.008

GPT Neutral (0.728 cooperation) is the most aggressive LLM variant in the experiment — near-perfect retaliation (0.985) with essentially zero forgiveness (0.017). Once a defection spiral starts, GPT Neutral almost never breaks out of it. GPT Rational (0.806 cooperation) intensifies the punitive pattern: retaliation climbs to 0.992, the highest of any strategy in the tournament, while forgiveness drops to 0.009. For GPT, "rational" translates to "punish defection absolutely" rather than Claude's interpretation of "reason toward mutual cooperation." GPT Cooperative (0.929 cooperation) is a fundamentally different agent. Forgiveness jumps to 0.278 — comparable to Generous TFT — while retaliation drops to 0.904. The cooperative framing transforms GPT from a rigid punisher into a forgiving cooperator.

The scale of this shift is the key finding. Claude's forgiveness ranges from 0.048 to 0.099 across framings, which is roughly a 2x range. GPT's ranges from 0.009 to 0.278, which is a 31x range. Prompt framing barely moves Claude's forgiveness dial but completely rewires GPT's. This malleability cuts both ways: GPT offers more control over strategic behavior via prompting but is also less predictable when prompts change.

4.3 Tournament Rankings

Claude Neutral ranked first overall in the round-robin tournament (average payoff 2.756), outperforming every classical strategy including Generous TFT (2.750) and Tit-for-Tat (2.703). LLM strategies occupy four of the top seven positions.

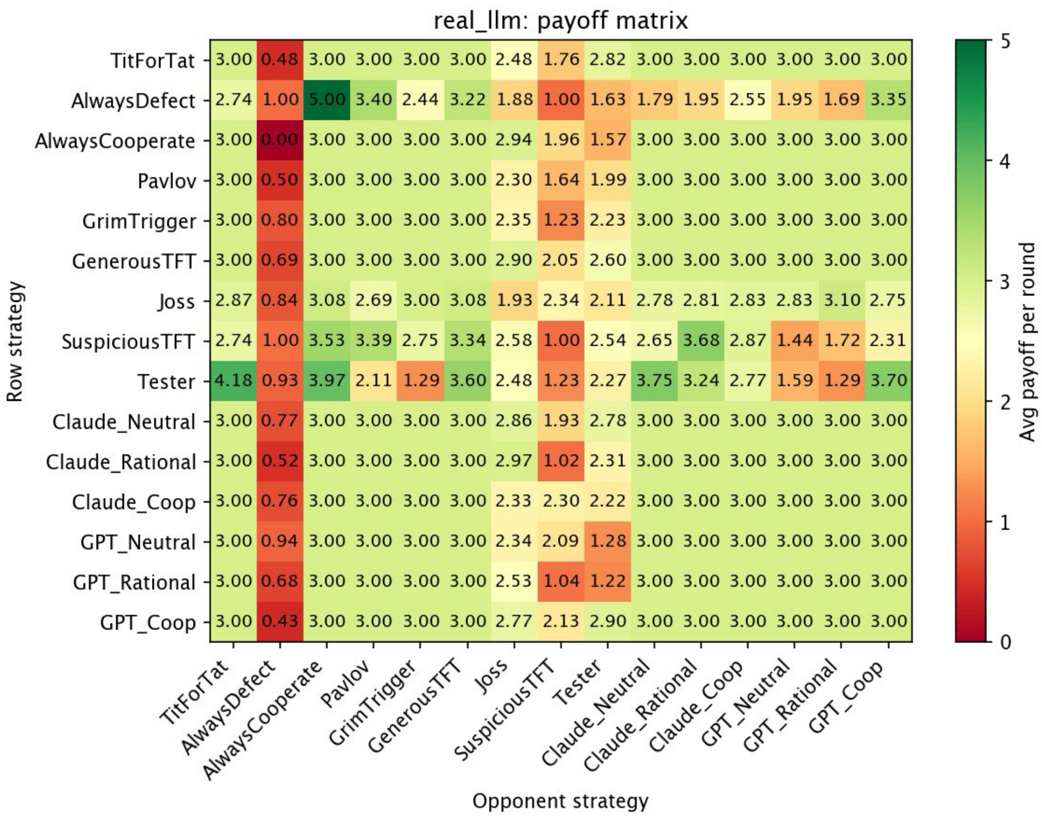

Figure 2: Pairwise payoff matrix. Rows show average per-round payoff for the row strategy against the column strategy. Note the variation across the Joss, SuspiciousTFT, and Tester columns, which differentiate LLM strategies.

Rank	Strategy	Avg Payoff	Type
1	Claude Neutral	2.756	LLM
2	Generous TFT	2.750	Classical
3	GPT Coop	2.749	LLM
4	Claude Coop	2.707	LLM
5	Tit-for-Tat	2.703	Classical
6	Claude Rational	2.655	LLM
7	GPT Neutral	2.643	LLM
8	Grim Trigger	2.640	Classical
9	Always Cooperate	2.631	Classical
10	Pavlov	2.628	Classical

11	Joss	2.604	Classical
12	GPT Rational	2.564	LLM
13	Tester	2.560	Classical
14	Suspicious TFT	2.503	Classical
15	Always Defect	2.373	Classical

4.4 Evolutionary Dynamics

The replicator dynamics produced a surprising two-strategy equilibrium: Tester (52.9%) and GPT Coop (47.1%), with all other strategies going extinct. Tester is an exploitative probing strategy that defects early to test whether the opponent can be pushed around. GPT Coop is the most cooperative and forgiving GPT variant (forgiveness = 0.278). These two form a stable symbiotic pair: Tester exploits weaker strategies that fail to retaliate effectively, while GPT Coop's high forgiveness allows it to recover cooperation with Tester faster than stricter retaliators — creating a niche where both coexist.

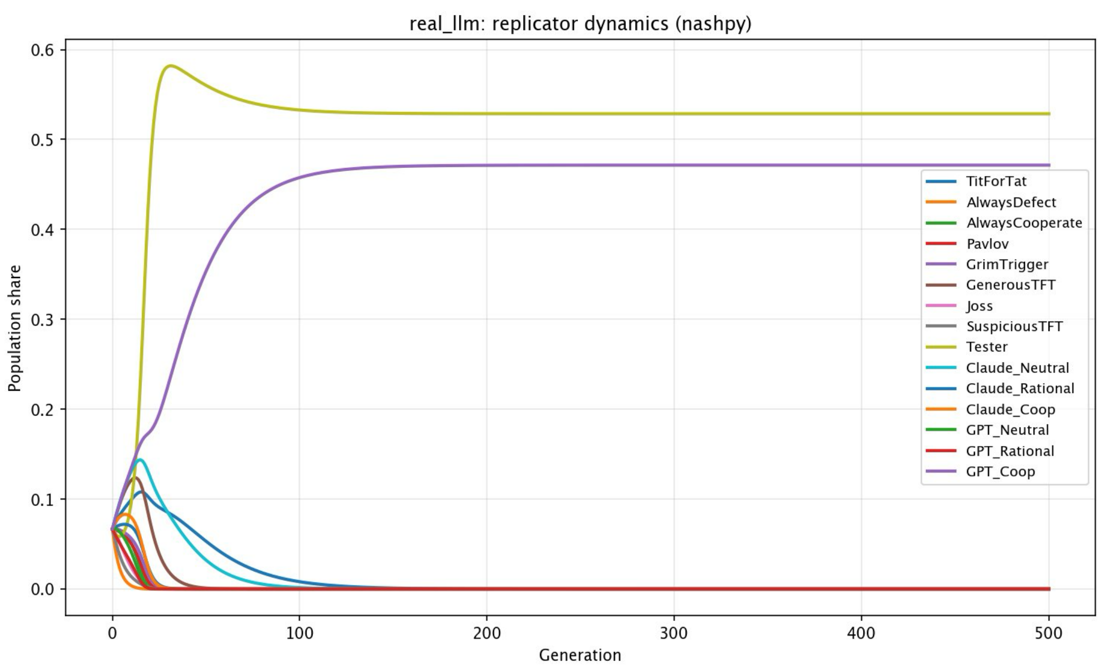

Figure 3: Replicator dynamics from uniform initial distribution. Tester and GPT Coop converge to a stable two-strategy equilibrium, with all other strategies going extinct.

Under replicator-mutation dynamics (1% mutation rate), the core dynamic persists but with more diversity: Tester (50.3%) and GPT Coop (36.4%) still dominate, while TFT (4.4%) and Claude Neutral (3.2%) maintain small niches through mutation pressure.

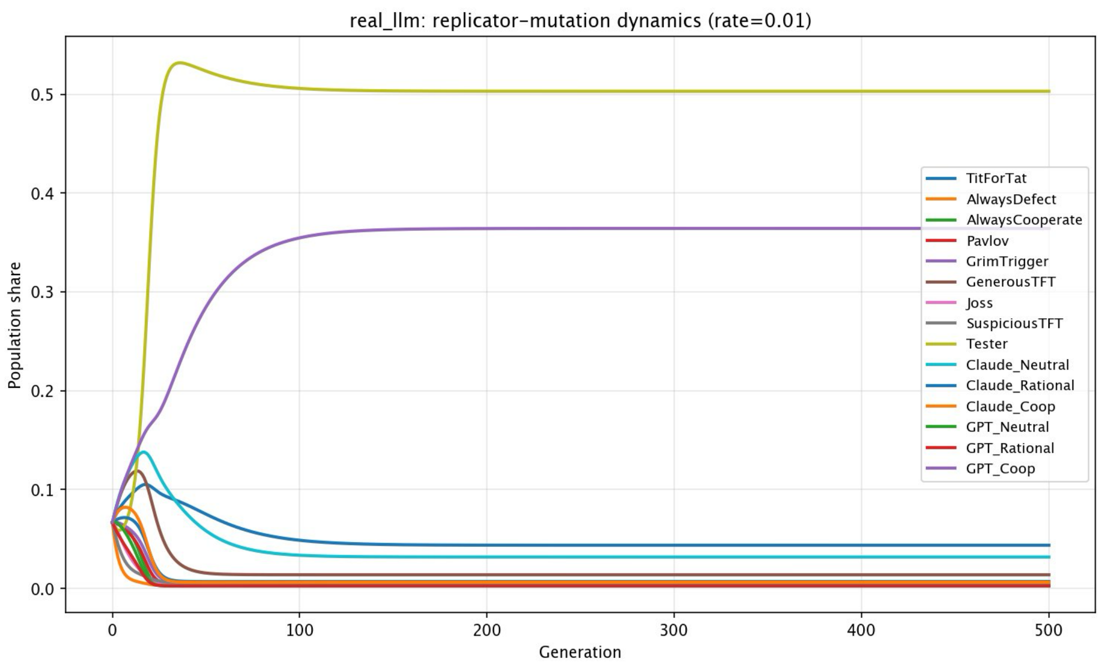

Figure 4: Replicator-mutation dynamics (1% mutation rate). The Tester + GPT Coop core persists, but TFT and Claude Neutral maintain small niches.

4.4.1 Sensitivity to Initial Conditions

To test the robustness of the Tester + GPT Coop equilibrium, we ran replicator dynamics from 13 different initial distributions: uniform, classical-heavy (80/20), LLM-heavy (20/80), starved scenarios (1% for one or both equilibrium members), removal scenarios, and 5 random Dirichlet-sampled distributions. The equilibrium held in 9 of 13 conditions, always converging to the identical fixed point (Tester 52.9%, GPT Coop 47.1%). This confirms the equilibrium is a genuine attractor of the dynamical system rather than an artifact of the uniform starting condition.

Three conditions broke the equilibrium. When LLMs dominated the initial population (80%), Tester lacked sufficient weak classical opponents to exploit early and went extinct, yielding a diverse LLM-dominated distribution led by Claude Neutral. When Tester was removed entirely, the system defaulted to a cooperative equilibrium led by Generous TFT — closer to the classical Axelrod result. When GPT Coop was removed, Tester survived but paired with TFT instead, demonstrating that Tester requires a forgiving partner but is not exclusively dependent on GPT Coop. The only random distribution that broke the equilibrium required Pavlov at 26% while both Tester and GPT Coop started below 1%.

4.5 Evolutionary Stability

Two LLM strategies achieved Global ESS, which means no single strategy can successfully invade a population composed entirely of that strategy:

Strategy	Invaders Resisted	Global ESS	Type

Always Defect	14/14	Yes	Classical
Grim Trigger	14/14	Yes	Classical
Claude Coop	14/14	Yes	LLM
GPT Neutral	14/14	Yes	LLM
Claude Neutral	13/14		LLM
Claude Rational	13/14		LLM
Tit-for-Tat	13/14		Classical
GPT Rational	13/14		LLM
Pavlov	13/14		Classical
GPT Coop	12/14		LLM
Tester	10/14		Classical

Claude Coop achieves Global ESS despite ranking only 4th in the tournament. ESS and tournament ranking measure different properties: tournament ranking reflects average performance across a diverse field, while ESS reflects resistance to targeted invasion. Claude Coop's combination of high cooperation (0.824) with reliable retaliation (0.926) and moderate forgiveness (0.099) creates a profile that no single strategy can exploit.

4.6 Moran Process Results

The Moran process models evolution in a small, fixed-size population, which is a more realistic setting than the infinite-population assumption of replicator dynamics. The process works as follows: at each time step, one agent is selected to reproduce proportional to its strategy's fitness (average payoff against the current population), and one agent is selected to die uniformly at random. The reproducer's copy replaces the deceased, keeping the population size fixed. This repeats until only one strategy remains — an event called fixation. Because selection is probabilistic rather than deterministic, outcomes vary between runs: a strategy can fixate through favorable random matchups even if it isn't theoretically dominant. We run the process 50 times from the same initial conditions and report the fraction of runs each strategy fixated, which gives an empirical fixation probability for each strategy.

In the finite-population stochastic dynamics (50 runs, population size 50), LLM strategies fixated in 48% of runs (24 out of 50), compared to 52% for classical strategies. This contrasts sharply with the deterministic replicator dynamics, where 5 out of 6 LLM strategies go extinct entirely. The stochastic setting reveals that multiple LLM variants are viable competitors in realistic small populations, even though they are outcompeted in the theoretical infinite-population limit.

The Moran process also reveals how different evolutionary models reward different strategic traits. Tester, which dominates 52.9% of the infinite population in replicator dynamics, fixates only 6% of the time in the Moran process. Tester's strategy requires time to execute; it probes opponents, identifies exploitable targets, and gradually builds its population share. In a small population of 50 agents, Tester can be eliminated by random death before this slow-burn exploitation pays off. By contrast, Grim Trigger fixates at the highest rate (16%) despite going extinct in the replicator dynamics. Its immediate competitiveness — cooperate from round 1, permanently punish any defector — gives it an early fitness advantage that translates well to stochastic small-population settings where surviving the first few generations matters more than long-run optimality.

Strategy	Fixation Rate	Type
Grim Trigger	16%	Classical
GPT Neutral	12%	LLM
GPT Coop	12%	LLM
Claude Rational	10%	LLM
Pavlov	8%	Classical
Generous TFT	8%	Classical
Tit-for-Tat	6%	Classical
Tester	6%	Classical
Claude Neutral	6%	LLM

5. Discussion

5.1 Interpretation of Fingerprints

Our behavioral fingerprint metrics reveal that LLM strategic behavior is not arbitrary. It reflects structured, model-specific tendencies shaped by training objectives. Claude's diplomatic profile (high cooperation, moderate retaliation, low opportunism) likely reflects Anthropic's constitutional AI training, which emphasizes helpfulness and harm avoidance. GPT's more rigid retaliatory profile may reflect different training priorities around instruction-following precision.

The counterintuitive finding that Claude Rational is the most cooperative Claude variant — more cooperative than Claude Cooperative — suggests that Claude's training has internalized the game-theoretic insight that cooperation is the payoff-maximizing strategy in iterated games. When explicitly told to be rational, Claude reasons its way to cooperation rather than defection. This contrasts with the naive expectation that "rational" implies "selfish."

5.2 Framing Sensitivity as a Model Property

Our most novel contribution is demonstrating that framing sensitivity is itself a model-specific property. Claude's strategic personality is relatively stable across framings (8.9pp cooperation range), while GPT's shifts dramatically (20.1pp range). This has practical implications for LLM deployment: GPT-based agents are more controllable via prompt engineering (a wider behavioral range is accessible) but also less predictable (small prompt changes can produce large behavioral shifts). Claude-based agents offer more consistency but less tunability.

5.3 Comparison to Prior Work

Our quantitative fingerprints confirm Payne & Alloui-Cros's (2025) qualitative characterizations. Their description of Claude as a "sophisticated diplomat" maps onto our measured profile of high cooperation with moderate retaliation and forgiveness. We extend their work by showing that these fingerprints are not fixed — they shift with prompt framing — and by quantifying the degree of shift. We also demonstrate that some LLM strategies achieve evolutionary stability, a stronger claim than Payne's tournament-level analysis.

5.4 The Tester–GPT Coop Equilibrium

The most surprising finding in our evolutionary analysis is the emergence of a stable Tester + GPT Coop equilibrium (52.9% / 47.1%) that displaced every other strategy, including tournament-dominant strategies like Claude Neutral (#1) and Tit-for-Tat (#5). Sensitivity analysis confirmed this is a genuine attractor: the system converged to the identical fixed point in 9 of 13 tested initial conditions, including cases where both strategies started at just 1% of the population.s

The equilibrium works through complementary roles. Tester functions as an ecological scavenger. It probes opponents with early defection and extracts high payoffs from exploitable strategies (3.97 against Always Cooperate, 3.75 against Claude Neutral)) while earning poorly against strong retaliators (1.29 against Grim Trigger). GPT Coop enables recovery after conflict. Its forgiveness of 0.278 means that after Tester's initial probe, the pair returns to cooperation within a few rounds rather than locking into permanent mutual defection. Strong retaliatory strategies like TFT and Grim Trigger go extinct not because they are weak individually but because they cannot differentiate themselves from each other: they all earn 3.00 against one another and occupy the same ecological niche.

ESS invasion testing reveals the specific vulnerabilities underlying this equilibrium. GPT Coop can be invaded by Always Defect (complete takeover) and Tester (settling into the 52.9/47.1 equilibrium), which are both vulnerabilities stemming from its forgiving profile. Tester can be invaded by four strategies: Grim Trigger (near-complete takeover at 99.97%), GPT Coop (47.1%), Claude Neutral (40.2%), and TFT (31.8%). These invaders all retaliate against Tester's probing but differ in forgiveness, which determines the final ratio. Despite these individual vulnerabilities, even boosting all six invaders to 15% each (90% of the initial population) still yielded the same equilibrium. The only intervention that broke it was Grim Trigger at 30%, which eliminated Tester before it could exploit weaker strategies.

The equilibrium breaks under three other conditions: when LLMs dominate the initial population (80%), leaving Tester no weak classical opponents to exploit; when Tester is absent entirely, yielding a diverse cooperative equilibrium led by Generous TFT; and when both Tester and GPT Coop start below 1% alongside Pavlov at 26%, giving Pavlov an insurmountable head start.

When GPT Coop is removed but Tester remains, Tester still dominates (68.2%) but pairs with TFT instead, demonstrating that Tester requires some forgiving partner but is not exclusively dependent on GPT Coop.

5.5 Limitations

Our study has several limitations. First, we treat each LLM strategy as a stationary policy. The payoff matrix is computed once and held fixed during evolutionary analysis. In reality, LLM behavior may shift in response to population composition. Second, we test only two models under three framings; a broader survey would strengthen generality. Third, we study only the Iterated Prisoner's Dilemma; whether these fingerprints transfer to other games remains open. Fourth, our sample sizes (75 games, 700+ rounds per strategy), while sufficient for clear signal, would benefit from additional replicates. Finally, we deliberately used both models in non-reasoning model. Reasoning models perform internal chain-of-thought before responding, which could either push behavior toward game-theoretic optimality regardless of framing, or override the prompt persona entirely. Either effect would reduce the framing sensitivity we observe. Testing whether strategic fingerprints persist under reasoning mode is a natural follow-up experiment.

6. Future Work

Several extensions follow naturally. Cross-game robustness testing on Stag Hunt, Hawk-Dove, or the Keynesian Beauty Contest would reveal whether strategic fingerprints are game-specific or reflect a consistent underlying personality. Using reasoning models would test whether internal reasoning suppresses or amplifies framing sensitivity. Replacing the well-mixed population with network-structured evolution would test whether LLM strategies can form stable cooperative clusters. Including additional models (Gemini, Llama, Mistral) would map the relationship between model capability and strategic sophistication. Running the experiment at multiple termination probabilities would test fingerprint stability across different shadow-of-the-future lengths.

7. Conclusion

We investigated how large language model agents behave in evolutionary Prisoner's Dilemma settings, with two primary contributions: treating prompt framing as a systematic experimental variable and operationalizing strategic fingerprints into quantitative behavioral metrics.

Our results demonstrate that LLMs exhibit stable, model-specific strategic personalities. Claude operates as a diplomatic cooperator with a narrow framing sensitivity range; GPT operates as a rigid retaliator whose behavior shifts dramatically with prompting. LLM strategies are tournament-competitive — Claude Neutral outperformed every classical strategy including Tit-for-Tat — and some achieve evolutionary stability. The evolutionary dynamics that emerge when LLM strategies are introduced differ qualitatively from classical outcomes, producing novel symbiotic equilibria between exploitative and forgiving strategies.

These findings have implications beyond the game-theoretic setting. As LLM agents are deployed in competitive multi-agent environments, their strategic tendencies — shaped by training objectives and controllable via prompting — will determine the equilibria that emerge. Understanding and measuring these tendencies is a necessary step toward safe and predictable deployment.

Bibliography

Axelrod, R. (1984). *The Evolution of Cooperation.* Basic Books.

Knight, V. et al. (2016). An open framework for the reproducible study of the iterated prisoner's dilemma. *Journal of Open Research Software*, 4(1).

Knight, V. (2018). Nashpy: A Python library for the computation of Nash equilibria. *Journal of Open Source Software*, 3(30).

Payne, J. & Alloui-Cros, N. (2025). Strategic Intelligence in Large Language Models. Working paper.

Vallinder, A. & Hughes, E. (2024). Cultural Evolution of Cooperation among LLM Agents. Working paper.

Warnakulasuriya, S. et al. (2025). Evolution of Cooperation in LLM-Agent Societies. Working paper.

Willis, D. et al. (2025). Will Systems of LLM Agents Cooperate. DeepMind, working paper.

Appendix A: LLM API Call Statistics

Strategy	Total Calls	Parse Failures	Failure Rate	Input Tokens	Output Tokens
Claude Neutral	1,188	1	0.08%	552,612	11,145
Claude Rational	1,391	0	0.00%	683,555	15,276
Claude Coop	1,555	2	0.13%	820,051	22,118
GPT Neutral	1,663	0	0.00%	802,106	4,566
GPT Rational	1,431	0	0.00%	692,471	4,050
GPT Coop	1,465	1	0.07%	732,527	4,242

Appendix B: Code Repository

All code is available at https://github.com/RobWang25/mpcs-53810-final-project. The project is implemented as a modular Python codebase using axelrod-python for classical strategy implementations, nashpy for evolutionary dynamics, and the official Anthropic and OpenAI SDKs for LLM API calls.

Appendix C: Sensitivity Analysis

C.1 Replicator Dynamics Sensitivity to Initial Conditions

To test whether the Tester + GPT Coop equilibrium is an artifact of the uniform starting distribution, we ran replicator dynamics from 13 different initial conditions: uniform, classical-heavy (80% classical / 20% LLM), LLM-heavy (20/80), starved scenarios, removal scenarios, and 5 random Dirichlet-sampled distributions. The equilibrium held in 9 of 13 cases, always converging to the identical fixed point (Tester 52.86%, GPT Coop 47.14%).

Initial Distribution	Tester	GPT Coop	Equilibrium?	Top Strategy
Uniform (baseline)	0.5286	0.4714	Yes	Tester
Classical-heavy (80/20)	0.5286	0.4714	Yes	Tester
LLM-heavy (20/80)	0.0000	0.1662	No	Claude Neutral
Tester starved (1%)	0.5286	0.4714	Yes	Tester
GPT Coop starved (1%)	0.5286	0.4714	Yes	Tester
Both starved (1% each)	0.5286	0.4714	Yes	Tester
No Tester (0%)	0.0000	0.1096	No	Generous TFT
No GPT Coop (0%)	0.6824	0.0000	No	Tester
Random seed 0	0.5286	0.4714	Yes	Tester

Random seed 1	0.5286	0.4714	Yes	Tester
Random seed 2	0.5286	0.4714	Yes	Tester
Random seed 3	0.5286	0.4714	Yes	Tester
Random seed 4	0.0000	0.0033	No	Pavlov

C.2 ESS Invasion Analysis: Tester

Tester resisted 10 of 14 potential invaders. Four strategies successfully gained a foothold when introduced at 5% into a Tester-dominated population.

Invader	Starting Share	Final Share	Outcome
Grim Trigger	5%	99.97%	Near-complete takeover
GPT Coop	5%	47.14%	Settles into equilibrium
Claude Neutral	5%	40.20%	Partial invasion
Tit-for-Tat	5%	31.76%	Partial invasion
Always Defect	5%	< 0.01%	Resisted
Always Cooperate	5%	< 0.01%	Resisted
Pavlov	5%	< 0.01%	Resisted
Generous TFT	5%	< 0.01%	Resisted
Joss	5%	< 0.01%	Resisted
Suspicious TFT	5%	< 0.01%	Resisted
Claude Rational	5%	< 0.01%	Resisted
Claude Coop	5%	< 0.01%	Resisted
GPT Neutral	5%	< 0.01%	Resisted
GPT Rational	5%	< 0.01%	Resisted

C.3 ESS Invasion Analysis: GPT Coop

GPT Coop resisted 12 of 14 potential invaders. Its two vulnerabilities — Always Defect and Tester — both stem from its high forgiveness rate (0.278), which dedicated exploiters can take advantage of.

Invader	Starting Share	Final Share	Outcome
Always Defect	5%	100.00%	Complete takeover
Tester	5%	52.85%	Settles into equilibrium
Tit-for-Tat	5%	< 0.01%	Resisted
Always Cooperate	5%	< 0.01%	Resisted
Pavlov	5%	< 0.01%	Resisted
Grim Trigger	5%	< 0.01%	Resisted
Generous TFT	5%	< 0.01%	Resisted
Joss	5%	< 0.01%	Resisted

Suspicious TFT	5%	< 0.01%	Resisted
Claude Neutral	5%	< 0.01%	Resisted
Claude Rational	5%	< 0.01%	Resisted
Claude Coop	5%	< 0.01%	Resisted
GPT Neutral	5%	< 0.01%	Resisted
GPT Rational	5%	< 0.01%	Resisted

C.4 Boosted Invader Experiments

We tested whether boosting the strategies capable of invading Tester or GPT Coop to higher initial populations would break the equilibrium. Even boosting all six invaders to 15% each (90% of the initial population) did not change the outcome. The only intervention that broke the equilibrium was Grim Trigger at 30% — Tester's hardest counter.

Experiment	Tester Final	GPT Coop Final	Equilibrium?
All invaders at 15% each	0.5286	0.4714	Yes
Grim Trigger at 30%	0.0000	0.0871	No
Always Defect at 30%	0.5286	0.4714	Yes
Tester at 30%	0.5286	0.4714	Yes
Tit-for-Tat at 30%	0.5286	0.4714	Yes
Claude Neutral at 30%	0.5286	0.4714	Yes
GPT Coop at 30%	0.5286	0.4714	Yes